AI & Cyber Security Interplay: The Good, the Bad & the Ugly

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Cyber Security Ecosystem

ARMS RACE!

Example

We managed WannaCry with damage

Total loss from attacks potentially at $T level soon!

Defense Landscape

- Defenders are disorganized
- Too much work at vision level
- Lacking execution
  - Stakeholders in 'babel tower'
  - Lack of technical consensus
  - Operational security is absent
  - Professionalization missing
  - Policy setting needs improvement
  - Certification push but not enough on labelling
  - Market dominancy questions
- Total revenue "only" at $133B

Attack Landscape

- Cybercriminals are organized
- Nation state cyberwarfare
- Increase in quantity and diversity
  - Ransomware
  - Target Attacks
  - DDoS
  - Espionage
  - Brain Attacks (phishing, fake news, disinformation)
- In military terms
  - **THEY HAVE THE INITIATIVE**

Explosion of the attack surface

- QUANTITATIVELY: Scale of a 5G infrastructure
- QUALTITATIVELY: Innovations change the architecture and increase the attack surface

But creates new conditions

Innovation

- FLOURISHING: Cloud, AI/ML, DLT, Quantum,
- BIGGER SOLUTIONS: 5G, IoT, ITS, Cyber currency,
- MARKETS/INDUSTRY: Digital Service Providers (ICT, OTT, Enterprises becoming Service Providers)

Both too slow and too fast (lack of ethics view) to adopt innovation

Gives conditions to ...

Very fast to adopt innovation against "us"; good intelligence of market and industry dynamics

Digitalization

- Transformative driving force
- Pre-requisite for our future societies

GRAPHICS CREDIT: ARNAUD TEDDAI

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Managing Risk is a Journey

Assets & Risks Discovery
What/Why need to be protected

Design

Organizational Roles & Responsibilities

Training

Awareness

Patching and update management

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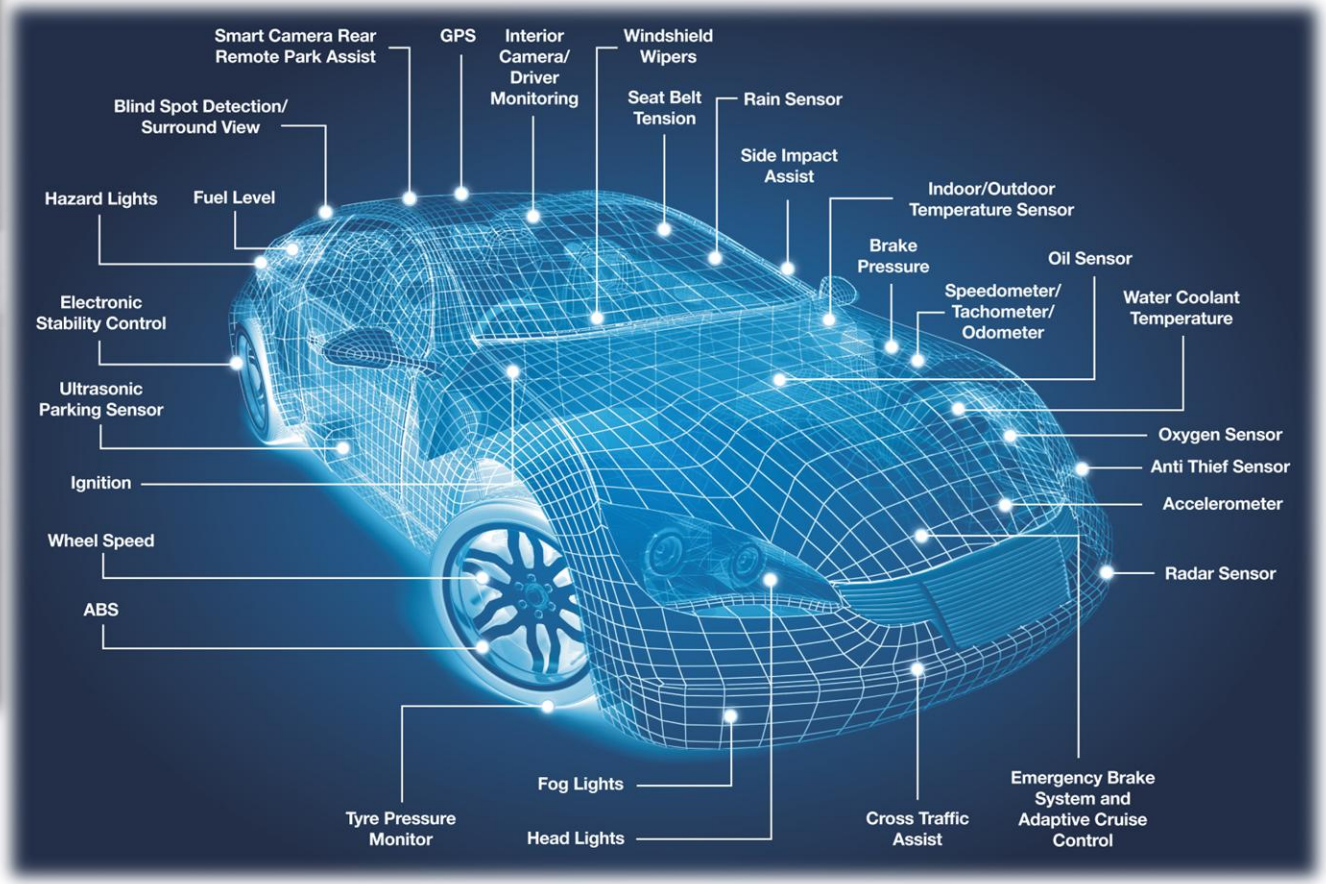
A Strategic Approach is Required

Tactical Risk Management

Strategic Risk Management



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Global Cyber Security Standardization

Direct impact to
- Regional/National strategy/priorities (E.g. EU CSA, NIS, GDPR, Data Spaces, AI, etc.)
- Certification/Labelling (e.g. ENISA)
- Regulation (e.g. Market Dominancy)
- Operation (e.g. Joint Cyber Unit, EU)

4 Stakeholders engaged in a huge battlefield

Administrations

Academia

Business

Civil Society

Stakeholders
"Dark Matter"

Frames

Structures =
Direct impact to
- Markets
- GDP
- Administrations

Security
Standardization
"Universe"

IEC ISO ITU

+ National Standard Bodies
Regional Standard Bodies, Industry Associations, etc.
NATO, MEF,

IETF
GSMA
3GPP

ETSI
OASIS
IEEE

BIS NIST

Coordination and collaboration exist but improvements are required

Security Standardization is increasingly fractalized

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Defending against Cyber Threats that think

- If 2025 provided us with AI hors d'oeuvre, 2026 will serve a full spread of Agents & Robots
- In 2026, cyber threats will be far more automated and scalable, driven by Agentic AI that can plan and execute entire attack campaigns with minimal human involvement.
- This shift fundamentally changes both the speed and complexity of modern cyber risk.
- Attackers are increasingly using AI systems to run reconnaissance, craft hyper-personalised phishing through deepfakes, and deploy polymorphic malware that constantly rewrites itself.
- They are hijacking legitimate automation workflows and manipulating AI assistants already embedded in enterprise environments.
- As a result, cybercrime is becoming more industrialised and harder for traditional defences to keep pace with.

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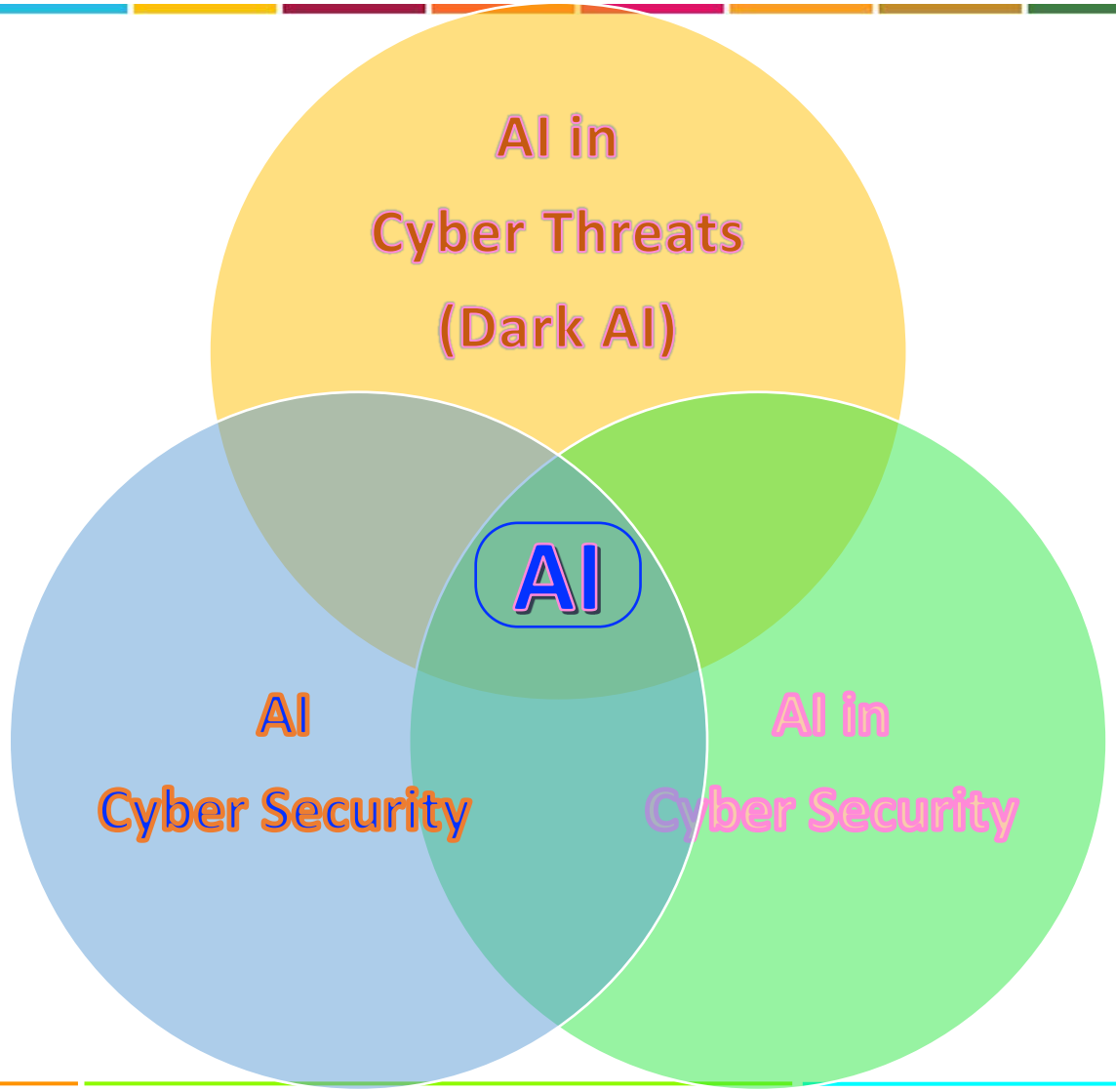
AI & Cyber Security Interplay...



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Interplay of AI and Cyber Security:

The Good, the Bad & the Ugly

- Artificial Intelligence (AI) has sparked widespread innovation and apprehension in the fields of technology, business, education, art, and more.
- There's no doubt in its role as one of the most powerful advancements of our time, or in its controversial applications.
- However, its influence is particularly relevant in the field of cybersecurity, where it impacts both defense and attack.
- AI is a double-edged sword. AI is transforming both sides of the cyber security chess match.
- For every defensive advance we make with AI-powered threat detection, attackers respond with AI-enhanced attacks and evasion techniques.
- The interplay between AI and cybersecurity is dynamic and multifaceted, involving both enhancements to security measures and new challenges introduced by AI technologies.
- The interplay creates a cybersecurity arms race where advancements in AI bolster defense capabilities but simultaneously empower threat actors with new tools.
- This arms race is accelerating at a pace we've never seen in the industry.
- The interplay of AI and cybersecurity is a complex and evolving landscape, encompassing positive advances, potential threats, and ethical challenges.

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The Good, the Bad & the Ugly

The Good:

- Enhanced Threat Detection: AI tools can identify vulnerabilities, detect anomalies, and predict cyber threats faster than traditional methods, improving overall security.
- Automated Response: AI systems can respond to incidents in real-time, containing attacks and minimizing damage.
- Improved Security Measures: Machine learning models help in developing smarter authentication systems, threat intelligence, and user behavior analytics.
- Proactive Defense: AI enables predictive cybersecurity, anticipating potential attacks before they happen.

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The Good, the Bad & the Ugly

The Bad:

- Evasion Techniques & Automation of Attacks: Use of AI enables rapid, automated attacks that can adapt in real-time, making threat mitigation more complex. Cybercriminals use AI to create more sophisticated attacks that can bypass traditional defenses.
- False Positives/Negatives: AI systems may generate false alarms or miss malicious activities, leading to resource wastage or security breaches.
- Data Poisoning: Attackers can manipulate training data of AI models, leading to compromised or biased AI systems.
- Skilled Adversaries: AI skills are accessible to malicious actors, increasing the risk of automated phishing, deepfakes, and social engineering.

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The Good, the Bad & the Ugly

The Ugly:

- Weaponization of AI: Autonomous hacking tools and AI-driven malware pose significant threats, potentially causing widespread damage.
- Fake & Deepfake: Malicious actors leverage AI to develop more sophisticated attacks, such as adaptive malware, deepfakes for social engineering, or evasive phishing campaigns.
- Ethical Concerns: Deploying AI in security raises concerns about surveillance, privacy violations, and misuse of personal data. Bias in AI models, and misuse of data can undermine trust and lead to societal harms.
- Arms Race: Continuous escalation between offensive AI techniques and defensive measures can destabilize cybersecurity efforts.

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AI in Cyber Security

- The security community has been using AI since the 1980s, but recent advancements have made it much more effective.
- There are several security use cases for AI including data security, identity and access management, IT management, cloud security, and threat detection and response.
- AI has transformed cybersecurity, making it easier for security professionals to respond to a growing number of cyberthreats.
- Future advancements in AI will continue to drive product development and new collaborations between people and AI-powered systems.

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AI in Cyber Security...

- AI in cybersecurity leverages artificial intelligence and machine learning to enhance threat detection, automation, and overall security posture.
- This includes analyzing vast amounts of data, identifying patterns, and responding to threats in real-time.
- AI-powered systems can learn from experience, predict future threats, and adapt to evolving tactics, making them more effective than traditional methods.
- By leveraging AI, organizations can create a more robust and adaptive cybersecurity posture, enabling them to effectively address the evolving landscape of cyber threats.

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AI in Cyber Security

- AI in cybersecurity refers to the use of AI technologies and techniques to enhance the protection of computer systems, networks, and data from cyberthreats.
- AI helps by automating threat detection, analyzing large volumes of data, identifying patterns, and responding to security incidents in real time.
- Key applications of AI for security include anomaly detection, malware detection, intrusion detection, fraud prevention, incident summaries, stakeholder reporting and building and reverse engineering scripts.
- By using machine learning, deep learning, and natural language processing, AI continuously learns from new data, improving its ability to identify and mitigate emerging threats, reduce false positives, and scale security efforts more effectively.
- Recent advancements in generative AI have empowered teams with data-driven insights, easy-to-produce reports, and step-by-step mitigation recommendations.

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AI in Cyber Security...

- **Advanced Threat Detection and Real-time Monitoring:**
  - AI algorithms can analyze data for unusual patterns and behaviours, enabling early threat detection.
  - By identifying anomalies, AI can help prevent security breaches and minimize potential damage.
  - AI-powered systems can also monitor networks and systems in real-time, providing a comprehensive view of security posture.
- **Automation and Incident Response:**
  - AI can automate tasks such as vulnerability assessments, threat hunting, and incident response.
  - This frees up security professionals to focus on more strategic tasks, like developing policies and improving security strategies.
  - AI-powered tools can also assist in incident response by providing actionable recommendations and automating mitigation efforts.

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AI in Cyber Security...

- **Vulnerability Management and Proactive Defense:**
  - AI can identify and prioritize vulnerabilities in systems and networks, allowing organizations to address the most critical issues first.
  - By proactively addressing vulnerabilities, AI helps prevent cyberattacks and reduces the risk of breaches.
  - AI can also be used to develop and implement proactive security measures, such as network segmentation and access control.
- **Phishing and Malware Detection:**
  - AI algorithms can analyze emails and messages for suspicious patterns, helping to detect phishing attempts and malicious links.
  - AI can also identify and block known and unknown malware, protecting systems from infection.
  - By automating phishing and malware detection, AI helps reduce the burden on security analysts and improves their ability to respond to threats.

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AI in Cyber Security...

Data Loss Prevention (DLP) and Data Access Monitoring:

- AI can monitor data access and identify suspicious activity, such as unauthorized access or data exfiltration.
- AI can also help identify and monitor shadow data, which may be at risk of exposure or misuse.
- By detecting and preventing data loss, AI helps organizations protect their sensitive information and comply with regulatory requirements.

AI-Powered Security Copilot:

- AI-driven assistants, like Microsoft Security Copilot, can help security teams investigate threats and manage risks in real-time.
- These tools can automate tasks, provide actionable insights, and assist in developing and implementing security solutions.
- By leveraging AI, security teams can improve their ability to defend against attacks at machine speed and scale.

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AI in Cyber Security

- Threat Detection: AI analyses Data in real-time to detect patterns & anomalies, identifying potential cyber threats.
- Incident Response: AI automates & speeds up response to attacks, suggesting remediation steps to mitigate damage.
- Vulnerability Management: AI identifies week points in networks allowing businesses to focus on securing critical vulnerabilities.
- Predictive Analytics: AI predicts future threats by analyzing historical data and patterns, enabling proactive defences.
- Endpoint Protection & Malware Detection: AI detects unusual behaviours and flags malware, including zero-day exploits.
- Fraud Detection & Prevention: AI analyzes transactions in real time, identifying suspicious activities to prevent fraud.
- Cyber Asset Attack Surface Management (CAASM): AI tracks & monitors where data is stored & transferred, enhancing network security.
- Cloud Security: AI secures cloud environments offering real-time protection against threats, crucial for remote and hybrid setups.

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The benefits of automating AI in Cyber Security

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AI Security

AI security risks are increasing as organizations integrate AI into their systems.

As AI adoption grows, the attack surface expands, leading to new vulnerabilities like prompt injection attacks and data leaks.

Shadow AI is a growing security concern.

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Key AI Security Threats:

Here’s are some key AI security threats:

- **Data Breaches and Data Manipulation:**
- **Adversarial Attacks**
- **Model Theft and Supply Chain Attacks**
- **Bias and Discrimination**
- **Privacy Concerns**
- **AI-Powered Cyberattacks**
- **Lack of Transparency and Explainability**
- **Dependence on AI**
- **Existential Risks**

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Key AI Security Threats:

Other AI Security Risks:

- Hardware vulnerabilities
- API attacks
- Prompt injection
- Membership inference attacks
- Data validation and security
- Model security enhancements
- Access controls
- Regular security audits
- Ethical AI practices
- Continuous monitoring and incident response

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DARK AI

- Unlike traditional AI — which can be applied to improve efficiency, enhance decision-making, or automate tasks — **dark AI** is specifically engineered to leverage these benefits to conduct cyberattacks, infiltrating systems and manipulating data.
- The core function of dark AI is to exploit vulnerabilities in digital infrastructures. The effects of dark AI often go unnoticed until significant damage is already done.
- While mimicking human behaviour through GenAI technologies, dark AI might create deceptive content or self-learn to achieve its objectives.
- By learning to bypass existing cybersecurity measures, dark AI becomes particularly dangerous.
- Because its evolving nature can outsmart conventional security defenses, dark AI complicates the detection and mitigation of cyber threats.
- It can also be used to automate sophisticated attacks at scale.
- This poses a threat to a wide range of information and digital systems, from personal data to critical infrastructure.
- The dark web has transformed over the last deca
- de from niche forums to complex marketplaces enabling a wide array of illegal activities. This shift is now amplified by AI-driven tools designed with malicious intentions like WormGPT, FraudGPT, and DarkBERT, which have recently been highlighted by the media.

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FraudGPT: a dark AI example

A real-world example of dark AI is FraudGPT, a tool designed for cybercriminal activities and sold on the dark web. FraudGPT is a GenAI tool with an interface similar to ChatGPT that is designed to enable threat actors to:

- Write malicious code
- Create undetectable malware
- Create phishing pages
- Create hacking tools
- Find leaks and vulnerabilities
- Write scam pages/letters

FraudGPT was discovered by cybersecurity researchers at Netenrich in July 2023. At the time, researchers saw the emergence of FraudGPT on Telegram channels and in dark web forums. The advertisement for FraudGPT on the dark web included a video of the tool at work. The researchers at Netenrich captured and posted a screenshot from that advertisement.

Critical Competencies needed...

For cybersecurity professionals this demands three critical competencies:

- First, adversarial AI understanding. Knowing how attackers poison models, exploit inference servers, and weaponise automation tools.
- Second, exposure-led prioritisation. With vulnerabilities surfacing at machine speed, engineers must focus on what truly increases business risk rather than chasing every alert.
- Third, defensive AI fluency, leveraging machine learning to detect anomalies and accelerate response while maintaining human judgement for critical decisions.
- The paradox is that AI is being used at scale in both attack and defence.
- To succeed, security professionals should be capable of operating at machine speed but retain the human judgement to know when context demands override.

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The Way Forward...

- While AI greatly enhances cybersecurity capabilities, it also introduces new vulnerabilities and ethical dilemmas.
- Effective cybersecurity strategies now increasingly rely on AI-driven solutions, but they must also incorporate measures to mitigate AI-related risks.
- Balancing innovation with responsibility is key to harnessing AI's potential for good while mitigating risks.

Strategic Approaches

- Adversarial Machine Learning Defense: Develop techniques to harden AI models against manipulation.
- Robust Data Governance: Ensure high-quality, secure training data to prevent poisoning.
- Continuous Monitoring and Updating: Regularly update AI models to adapt to evolving threats.
- Ethical AI Use: Incorporate privacy-preserving techniques and transparency to maintain trust.

Cyber Immunity & Cyber Resilience

An intuitive and adaptive cyber posture defined by zero latency networks and quantum leaps will be needed across industries.

Cyber Immunity at every layer will create networks that are inherently secure and self-learning.

- AI-induced *digital intuition* is one of the pillars of cyber-Security strategy that will allow intelligent adaption.
- The ability of AI systems to out-innovate malicious attacks by *mimicking various aspects of human immunity* will be the line of defence to attain cyber resilience based on both supervised and unsupervised machine learning.
- The systems can be designed to make the right decisions with the context-based data, pre-empt attacks on the basis of initial indicators of compromise or attack, and *take intuitive remediated measures,* allowing any digital infrastructure and organization to be more Resilient.

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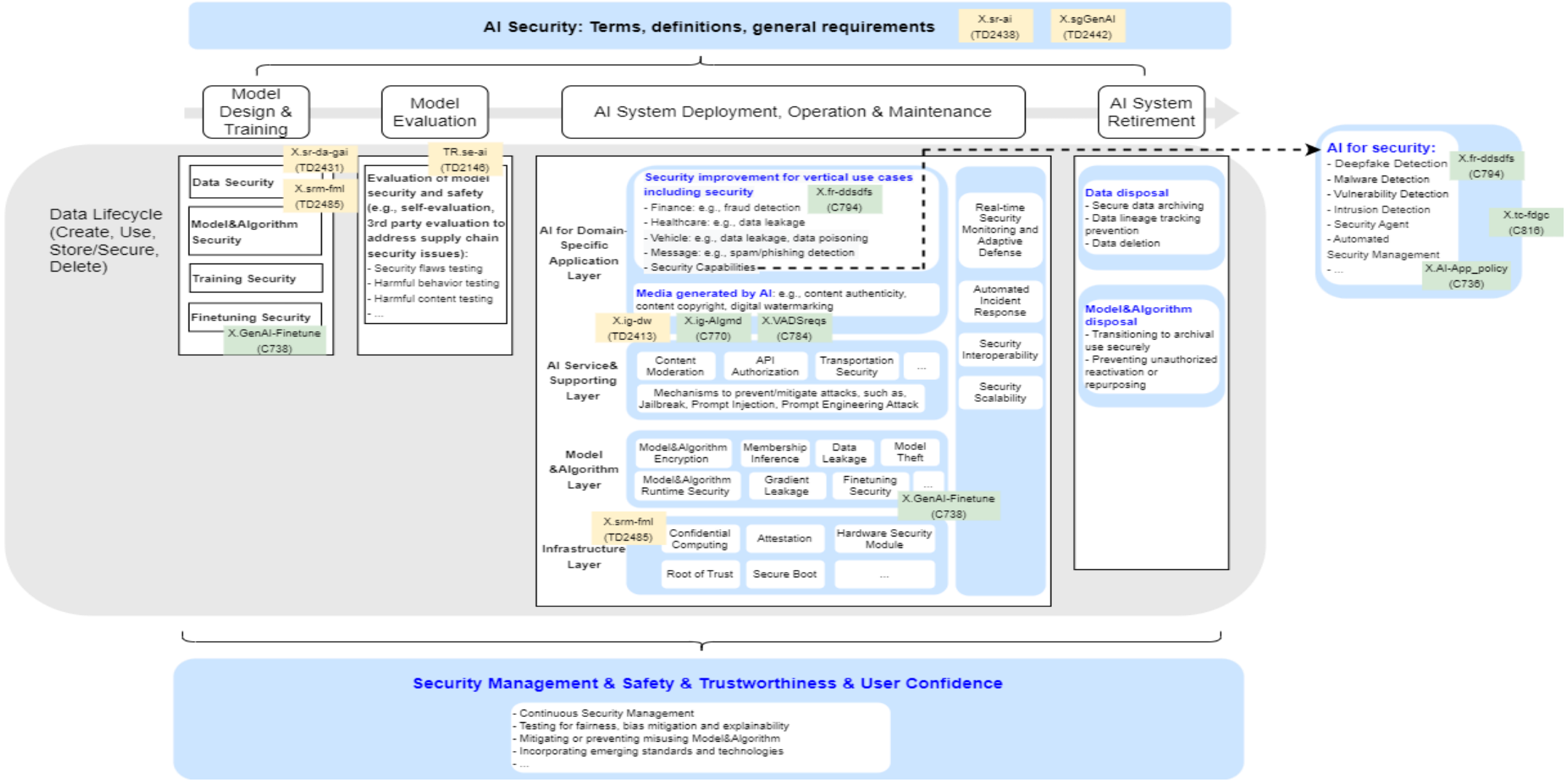
General Framework for AI Security



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THE POLICY CHALLENGE - CALL FOR ACTION

Stakeholders must be identified, including the relative allocation of threats/harms and promises/benefits among them

- SDOs
- Governments, Policy Makers & Regulators
- System Developers & Business Organizations
- Academia & Researchers
- Intergovernmental & NGOs

Ethical Imperatives for the Society...

We humans need to radically change our relationship, not just with the planet, but with the objects with which we fill our lives.

We need to change how we think about technology and innovation.

Rather than allowing technological advancement to steer our narratives, innovation and technology should help us build bridges between the worlds we inhabit now and the ones we imagine for tomorrow.

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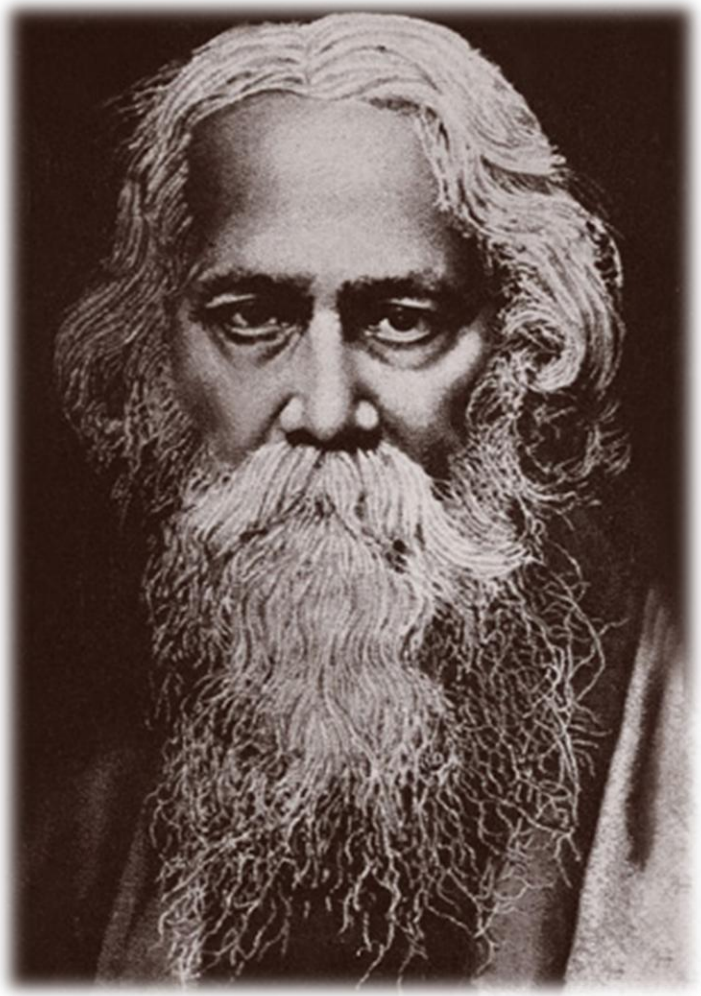
Resilience....



Let us not pray to be sheltered from dangers but to be fearless when facing them

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Thank you!

For a Sustainable & Resilient Future

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RACE TO ZERO

RACE TO RESILIENCE

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About me… N. KISHOR NARANG

Technology Philanthropist, Ethicist, Innovation, Standardization & Sustainability Evangelist…

Technology Advisor, Mentor & Design Strategist & Architect in Electrical, Electronics & ICT; running an Independent Design House – NARNIX since 1981

- Over 47 years of professional experience in education, research, design and advisory .
- Over 36 years of hardcore Research and Design Development Experience in Solutions, Systems, Products - Hardware, Software & Firmware (Embedded Software) in fields of Industrial, Power, IT, Telecom, Medical, Automotive, Aerospace, Defense, Energy and Environment. Over 10 years of Strategic Advisory Experience to different segments of business & industry.
- Over 500 Research & Design Mentees in the Electronics & ICT Ecosystems.
- **Mentoring many Deep Tech & Disruptive Tech Startups.**
- Leading & contributing to multiple National & Global Standardization Initiatives at BIS, Niti Aayog, TSDSI, IEC, ISO, ITU, IEEE etc….
- For the last 15 years, been deeply involved in standardization in the electrical, electronics, communications, information technology, digital infrastructure and cyber security domains with a focus on identifying gaps in standards to bring harmonization through system standards and standardized interfaces to ensure end-to-end Interoperability.
- **National Standards (INDIA) based on 10 years of Pre-Standardization Research Published Recently (December 2020) -**
  - Unified Digital Infrastructure ICT Reference Architecture – IS 18000:2020.
  - Unified Last Mile Communication Protocol Stack Reference Architecture – IS 18010:2020.

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- **Founding Chairman of the Smart Infrastructure Sectional Committee LITD 28 in BIS** (20 + standards on Unified Digital Infrastructure already published).
- **Vice Chair-Strategy, and Project Leader of two international standards in IEC SyC Smart Cities** – SCRAM: Smart Cities Reference Architecture Methodology IEC 63188 (published in September 2022) and SCRA: Smart Cities Reference Architecture IEC 63205 being developed jointly with ISO TC 268 & ITU-T SG 20.
- **Member IEEE SA Standards Board**
- **Chair, IEEE SA Working Group P2784 - Smart Cities Technology Framework.**
- Vice Chair, IEEE SA Working Group P 2872 - Standard for Interoperable and Secure Wireless Local Area Network (WLAN) Infrastructure and Architecture
- Convenor, Communication Technologies Work Group in IEC SyC Communication Technologies & Architectures,
- Convenor, IEC SyC SET, ahG 1: Systems Architecture for Sustainable Electrified Transportation
- Convenor, IEC TC 13/JahG 17 – Identifying Synergy & Interplay between TC 13 & TC 69
- Member, IEC SMB/SG 12 -Digital Transformation and Systems Approach,
- Member, IEC SMB/SG 14– All Electrified & Connected Society
- Member, IEC SMB/ahG 96 - Governance of Artificial Intelligence
- Member, IEC MSB/SWG 14 - SMART Standards – A Market & Industry perspective.
- OCEANIS - Chair of Advancing Research Work Group & member Steering Committee.
- ITU-T SG 20, ITU-T SG 17 & ITU-R SG 5 - Proactive Contributor.
- ITU-T SG 17 – Editor – X.cs-ra – Cyber Security Reference Architecture; Co-editor, X.arch-design: Design principles and best practices for security architectures.
- Member UL STP 3600 - ‘Measuring and Reporting Circular Economy Aspects of Products, Sites and Organizations’.
- ISO/IEC JTC1/WG11 - Co-Editor in Four Standards.
- **IEEE IoT Magazine** - Standards Column Editor.